



From the D ogram Wiki

The many **upgrades** in Dyson Sphere Program can be used throughout the game to improve Icarus's abilities, speed up production lines, optimize logistics, and increase the Dyson sphere's effectiveness. Below are the upgrades and their appropriate costs.

Completing all upgrades (exluding repeatables and those requiring items other than matricies) requires:



Note: Upgrades are sorted by how they appear from top to bottom and by tier.

Exploration Upgrades

These upgrades highlight where resources are located on a planet and eventually show the player what resources are located in other star systems. As such, these are mostly quality-of-life upgrades.

Tier	Icon	Upgrade	ID	Resource Consumption				Unlocks	Notes
1		Cosmic Exploration 1	4101	10				Viewing capabilities for vein distribution on current planet.	
2		Cosmic Exploration 2	4102	200				Viewing capabilities for vein distribution on current planetary system.	
3		Cosmic Exploration 3	4103	1000	1000	1000		Viewing capabilities for vein distribution on planetary systems within 6 light-years.	Requires Drive Engine 2. The range refers to within 6 light-years of Icarus, not 6 light-years from the home system.
4		Cosmic Exploration 4	4104	2000	2000	2000	2000	Viewing capabilities for vein distribution and rare veins on all planetary systems.	

Mecha Upgrades

These upgrades improve Icarus's abilities, ranging from walking and building speed to interplanetary and interstellar space flight.

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
1		Mecha Core 1	2101	20	20				+80MJ maximum core energy limit. +300 Mecha Durability +1m Mecha laser weapon range	Max energy: +80MJ (200MJ). Durability: +300 (1800) Range: +1m (11m)	
2		Mecha Core 2	2102	100	100				+200MJ maximum core energy limit. +300 Mecha durability +1m mecha laser weapon range +2hp Mecha laser weapon damage	Max energy: +280MJ (400MJ). Durability: +600 (2100) Range: +2m (12m) Damage: +2hp (28)	
3		Mecha Core 3	2103	300	300	300			+400MJ maximum core energy limit. +300 Mecha durability +2m mecha laser weapon range +2hp Mecha laser weapon damage	Max energy: +680MJ (800MJ). Durability: +900 (2400) Range: +4m (14m) Damage: +4hp (30)	
4		Mecha Core 4	2104	500	500	500	500		+800MJ maximum core energy limit. +300 Mecha durability +2m mecha laser weapon range +2hp Mecha laser weapon damage	Max energy: +1.48GJ (1.6GJ). Durability: +1200 (2700) Range: +6m (16m) Damage: +6hp (32)	
5		Mecha Core 5	2105	800	800	800	800	800	+1.6GJ maximum core energy limit. +300 Mecha durability +2m mecha laser weapon range +2hp Mecha laser weapon damage	Max energy: +3.08GJ (3.2GJ). Durability: +1500 (3000) Range: +8m (18m) Damage: +8hp (34)	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
6		Mecha Core ∞	2106	1000 × N*(N+1)/2; N=LVL-5	+2.4GJ maximum core energy limit. +200 Mecha durability +2hp Mecha laser weapon damage	Max energy: + (3.08+2.4*(LVL-5)) GJ. Total: (3.2+2.4*(LVL-5)) GJ Durability: + (1500+200*(LVL-5)) Total: (3000+200*(LVL-5)) Range: +8m (18m) Damage: + (8+2*(LVL-5)) hp Total: (34+2*(LVL-5)) hp	Can be researc infinitely.
1		Mechanical Frame 1	2201	60	+1 m/s mecha move speed.	Move speed: +1 m/s (7 m/s).	Requires Mecl Core 1.
2		Mechanical Frame 2	2202	200, 200	+1 m/s mecha move speed.	Move speed: +2 m/s (8 m/s).	
3		Mechanical Frame 3	2203	400, 400	+1 m/s mecha move speed.	Move speed: +3 m/s (9 m/s).	Requires Mecl Core 3.
4		Mechanical Frame 4	2204	600, 600, 600	+1 m/s mecha move speed.	Move speed: +4 m/s (10 m/s).	
5		Mechanical Frame 5	2205	800, 800, 800	+1 m/s mecha move speed.	Move speed: +5 m/s (11 m/s).	Requires Mecl Core 3
6		Mechanical Frame 6	2206	1000, 1000, 1000, 1000	+1 m/s mecha move speed.	Move speed: +6 m/s (12 m/s).	Requires Mecl Core 4.
7		Mechanical Frame 7	2207	1200, 1200, 1200, 1200	+2 m/s mecha move speed.	Move speed: +8 m/s (14 m/s).	
8		Mechanical Frame 8	2208	2000, 2000, 2000, 2000, 2000	+2 m/s mecha move speed.	Move speed: +10 m/s (16 m/s).	Requires Mecl Core 5.
1		Inventory Capacity 1	2301	120, 120	+1 Line(s) inventory capacity	+1 line (5 = 50 slots).	Requires Mecl Core 1.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
2		Inventory Capacity 2	2302	200 200	+1 Line(s) inventory capacity	+2 lines (6 = 60 slots).	
3		Inventory Capacity 3	2303	400 400 400	+1 Line(s) inventory capacity +2 Logistics List stack multiplier	+3 lines (7 = 70 slots).	Requires Mecl Core 2.
4		Inventory Capacity 4	2304	600 600 600	+1 Line(s) inventory capacity +1 column(s) on Logistics List +2 Logistics List stack multiplier	+4 lines (8 = 80 slots). Logistics List columns: 2	
5		Inventory Capacity 5	2305	800 800 800	+2 Line(s) inventory capacity +2 Logistics List stack multiplier	+6 lines (10 = 100 slots).	Requires Mecl Core 3.
6		Inventory Capacity 6	2306	1000 1000 1000 1000 1000	+2 Line(s) inventory capacity +2 Logistics List stack multiplier	+8 lines (12 = 120 slots).	Requires Mecl Core 5.
7		Inventory Capacity 7	2307	1500	+2 Logistics List stack multiplier +1 column(s) on Logistics List	+8 lines (12 = 120 slots). Logistics List columns: 3	
1		Communication Control 1	2401	100	+1 construction drone.	+1 drone (4).	Requires Mecl Core 1.
2		Communication Control 2	2402	200 200	+2 construction drone.	+3 drones (6).	
3		Communication Control 3	2403	400 400	+1 construction drone task.	+3 drones (6). +1 drone task (2).	Requires Mecl Core 2.
4		Communication Control 4	2404	600 600 600	+3 construction drones.	+6 drones (9). +1 drone task (2).	Requires Mecl Core 3.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
5		Communication Control 5	2405	1000 1000 1000 1000	+3 construction drones.	+9 drones (12). +1 drone task (2).	Requires Mecl Core 4.
6		Communication Control 6	2406	1600 1600 1600 1600 1600	+2 construction drone tasks.	+9 drones (12). +3 drone tasks (4).	Requires Mecl Core 5.
7		Communication Control ∞	2407	2000 × (LVL-6)^2	+3 construction drones.	+ (9+3*(LVL-6)) drones. Total drones: (12+3*(LVL-6)).	Can be researc up to level 88 (Number of dr capped to 256
1		Auto Reconstruction Marking 1[1]	2951	200	Auto Reconstruction Marking: 1/s		Requires Mecl Core 1.
2		Auto Reconstruction Marking 2	2952	400 400	Auto Reconstruction Marking: 3/s		
3		Auto Reconstruction Marking 3	2953	800 800 800	Auto Reconstruction Marking: 6/s		
4		Auto Reconstruction Marking 4	2954	1200 1200 1200 1200	Auto Reconstruction Marking: 15/s		
5		Auto Reconstruction Marking 5	2955	1600 1600 1600 1600 1600	Auto Reconstruction Marking: 30/s		
6		Auto Reconstruction Marking 6	2956	3200	Quick-mark targets to be reconstructed		
1		Mass Construction 1	2701	100	Blueprint Limit: 150 Facilities. Blueprint Limit: 1200 Foundations.		Requires Mecl Core 1. Requires reacl 6. Also unlocks t Blueprint featu
2		Mass Construction 2	2702	300	Blueprint Limit: 300 Facilities. Blueprint Limit: 2400 Foundations.		Requires reacl 30.
3		Mass Construction 3	2703	500 500	Blueprint Limit: 900 Facilities. Blueprint		Requires Mecl Core 2.

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
									Limit: 7200 Foundations.		Requires reacl 30.
4		Mass Construction 4	2704	800	800	800			Blueprint Limit: 3600 Facilities. Blueprint Limit: 28800 Foundations.		Requires Mecl Core 3. Requires reacl 30.
5		Mass Construction 5	2705	1000	1000	1000	1000		No facility quantity limit on Blueprint. Blueprint Limit: 45000 Foundations.		Requires Mecl Core 4. Requires reacl 30.
6		Mass Construction 6	2706	2000	2000	2000	2000	2000	No foundation quantity limit on Blueprint.		Requires Mecl Core 5. Requires reacl 30.
1		Energy Circuit 1	2501	60	60	60			+200kW mecha fuel power.	Fuel power: +200kW (1MW).	Requires Mecl Core 1.
2		Energy Circuit 2	2502	400	400				+200kW mecha fuel power.	Fuel power: +400kW (1.2MW).	Requires Mecl Core 2.
3		Energy Circuit 3	2503	1200	1200	1200			+200kW mecha fuel power.	Fuel power: +600kW (1.4MW).	Requires Mecl Core 3.
4		Energy Circuit 4	2504	2000	2000	2000			+200kW mecha fuel power.	Fuel power: +800kW (1.6MW).	Requires Mecl Core 4.
5		Energy Circuit 5	2505	3000	3000	3000	3000	3000	+800kW mecha fuel power.	Fuel power: +1.6MW (2.4MW).	Requires Mecl Core 5.
6		Energy Circuit ∞	2506	4000 × N*(N+1)/2; N=LVL-5					+1MW mecha fuel power.	Fuel power: +(1.6+1*(LVL-5))MW. Total: (2.4+1*(LVL-5))MW	Can be researc infinitely.
1		Drone Engine 1	2601	200					+2 m/s construction drone flight speed	Drone speed: +2 m/s (8 m/s)	Requires Mecl Core 1.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
2		Drone Engine 2	2602	400 400	+2 m/s construction drone flight speed	Drone speed: +4 m/s (10 m/s)	Requires Mecl Core 2.
3		Drone Engine 3	2603	800 800 800	+2 m/s construction drone flight speed	Drone speed: +6 m/s (12 m/s)	Requires Mecl Core 3.
4		Drone Engine 4	2604	1200 1200 1200 1200	+3 m/s construction drone flight speed	Drone speed: +9 m/s (15 m/s)	Requires Mecl Core 4.
5		Drone Engine 5	2605	1600 1600 1600 1600 1600	+3 m/s construction drone flight speed	Drone speed: +12 m/s (18 m/s)	Requires Mecl Core 5.
6		Drone Engine ∞	2606	2000 × (LVL-5)^2	+3 m/s construction drone flight speed	Drone speed: +(15+3*(LVL-5)) m/s. Total: (18+3*(LVL-5)) m/s.	Can be researc up to LVL 24 (Drone Speed to 75 m/s).
1		Drive Engine 1	2901	150 50	Unlocks mecha flying ability. It's faster than walking, but consumes energy faster.		Requires Mecl Core 1.
2		Drive Engine 2	2902	200 200	Unlocks mecha sail ability. Once a high enough altitude is reached, the mech can leave the atmosphere and travel to other planets. Consumes energy even faster than flying.	Maximum Sail Speed: 1000 m/s	Requires Mecl Core 2.
3		Drive Engine 3	2903	1000 1000 1000	+1000 m/s sail speed.	Maximum Sail Speed: 2000 m/s	Requires Mecl Core 3.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
4		Drive Engine 4	2904	2000 2000 2000	Unlocks ability to warp across large distances using space warpers	Warp Speed: 12 AU/s	Requires Gravitational W Refraction, Me Core 4.
5		Drive Engine 5	2905	3000 3000 3000 3000	+3 AU/s Mecha Warp Speed	Warp Speed: +3 AU/s (15 AU/s)	
6		Drive Engine ∞	2906	1000 × (LVL*(LVL-5))	+3 AU/s Mecha Warp Speed	Warp Speed: +3*(LVL-5) AU/s (15.5+3*(LVL-5) AU/s)	Can be researc up to LVL 100 (Warp Speed u 5 LY/s).
1		Energy Shield 1[1]	2801	100	Unlock Mecha Energy Shield Base Radius +2m Energy Shield Capacity +100 MJ	Radius: 2m Capacity: 100 MJ	Requires Electromagnet
2		Energy Shield 2	2802	100 100	Base Radius +0.5m Energy Shield Capacity +200 MJ	Radius: 2.5m Capacity: 300 MJ	Requires Mech Core 2.
3		Energy Shield 3	2803	200 200 200	Base Radius +0.5m Energy Shield Capacity +200 MJ	Radius: 3m Capacity: 500 MJ	Requires Mech Core 3.
4		Energy Shield 4	2804	600 600 600 600	Base Radius +1m Energy Shield Capacity +200 MJ	Radius: 4m Capacity: 700 MJ	
5		Energy Shield 5	2805	1000 1000 1000 1000 1000	Base Radius +1m Energy Shield Capacity +300 MJ	Radius: 5m Capacity: 1 GJ	Requires Mech Core 4.
6		Energy Shield 6-25	2806	1000 × (LVL-4)*(LVL-5)	Base Radius +1m Energy Shield Capacity +500 MJ	Radius: 1*LVL m Capacity: 0.5*(LVL-3) GJ	Requires Mech Core 5.
26		Energy Shield ∞	2807	1000 × (LVL-4)*(LVL-5)	Energy Shield Capacity +500 MJ	Radius: 25 m Capacity:	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
						`0.5*(LVL-3)` GJ	

Building Upgrades

These upgrades improve the efficiency of Splitters, Matrix Labs, Solar Sails, and more.

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
1		Solar Sail Life 1	3101	400	400	400			+300s solar sail life.	+300s solar sail life (5,700s).	Requires Solar Orbit System.
2		Solar Sail Life 2	3102	600	600	600			+300s solar sail life.	+600s solar sail life (6,000s).	
3		Solar Sail Life 3	3103	800	800	800			+600s solar sail life.	+1,200s solar sail life (6,600s).	
4		Solar Sail Life 4	3104	1000	1000	1000	1000		+600s solar sail life.	+1,800s solar sail life (7,200s).	
5		Solar Sail Life 5	3105	1600	1600	1600	1600		+900s solar sail life.	+2,700s solar sail life (8,100s).	
6		Solar Sail Life 6	3106	2000	2000	2000	2000		+900s solar sail life.	+3,600s solar sail life (9,000s).	
1		Solar Sail Attaching Speed 1	4201	600	600	600	600	600	Solar Sail Attaching Speed: 15 / min		Requires Vertical Launching Silo Starting value: min.
2		Solar Sail Attaching Speed 2	4202	1000	1000	1000	1000	1000	Solar Sail Attaching Speed: 30 / min		
3		Solar Sail Attaching Speed 3	4203	2500	2500	2500	2500	2500	Solar Sail Attaching Speed: 60 / min		
4		Solar Sail Attaching Speed 4	4204	4000	4000	4000	4000	4000	Solar Sail Attaching Speed: 120 / min		
5		Solar Sail Attaching Speed 5	4205	8000	8000	8000	8000	8000	Solar Sail Attaching Speed: 240 / min		
6		Solar Sail Attaching Speed 6	4206	12.0k	12.0k	12.0k	12.0k	12.0k	Solar Sail Attaching Speed: 360 / min		
1		Ray Transmission Efficiency 1	3201	600	600	600			-10% solar ray energy dissipation.	-7% solar ray energy dissipation (63%).	Requires Ray Receiver. Base solar ray energy dissipa is 70%. Cumulative eff

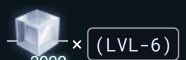
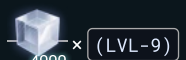
Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
											values listed a additive to tha value.
2		Ray Transmission Efficiency 2	3202	800	800	800			-10% solar ray energy dissipation.	-13.3% solar ray energy dissipation (56.7%).	
3		Ray Transmission Efficiency 3	3203	1000	1000	1000			-10% solar ray energy dissipation.	-18.97% solar ray energy dissipation (≈45.93%).	
4		Ray Transmission Efficiency 4	3204	1200	1200	1200	1200		-10% solar ray energy dissipation.	-28.66% solar ray energy dissipation (≈41.33%).	
5		Ray Transmission Efficiency 5	3205	1400	1400	1400	1400		-10% solar ray energy dissipation.	-32.8% solar ray energy dissipation (≈37.2%).	
6		Ray Transmission Efficiency 6	3206	1600	1600	1600	1600	1600	-10% solar ray energy dissipation.	-38.6% solar ray energy dissipation (≈31.62%).	
7		Ray Transmission Efficiency 7	3207	1800	1800	1800	1800	1800	-15% solar ray energy dissipation.	-43.12% solar ray energy dissipation (≈26.88%).	
8		Ray Transmission Efficiency ∞	3208	2000 × (LVL-7)					-15% solar ray energy dissipation.	Total: ≈ 26.88*(0.85^(LVL-7))%	Can be researc infinitely.
1		Vertical Construction 1	3701		400				+1 layer storage capacity +2 layers lab vertical construction.	+1 layer storage capacity (3) +2 layers lab vertical construction (5).	Splitters layer capacity equal that of storage Requires Basic Logistics Syste
2		Vertical Construction 2	3702		600				+1 layer storage capacity +2 layers lab vertical construction.	+2 layers storage capacity (4) +4 layers lab vertical construction (7).	

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
3		Vertical Construction 3	3703		800	800			+1 layer storage capacity +2 layers lab vertical construction.	+3 layers storage capacity (5) +6 layers lab vertical construction (9).	
4		Vertical Construction 4	3704	1000	1000	1000			+1 layer storage capacity +2 layers lab vertical construction.	+4 layers storage capacity (6) +8 layers lab vertical construction (11).	
5		Vertical Construction 5	3705	1600	1600	1600			+1 layer storage capacity +2 layers lab vertical construction.	+5 layers storage capacity (7) +10 layers lab vertical construction (13).	
6		Vertical Construction 6	3706	2000	2000	2000		2000	+1 layer storage capacity +2 layers lab vertical construction.	+6 layers storage capacity (8) +12 layers lab vertical construction (15).	
1		Pile Sorter Upgrade 1	3301	500	500	500			Pile Sorter feeds out: 2.		Requires Integrated Logistics Syste
2		Pile Sorter Upgrade 2	3302	800	800	800			Pile Sorter loads: 3.		
3		Pile Sorter Upgrade 3	3303	1000	1000	1000			Pile Sorter feeds out: 3.		
4		Pile Sorter Upgrade 4	3304	1200	1200	1200			Pile Sorter loads: 4.		
5		Pile Sorter Upgrade 5	3305	1500	1500	1500			Pile Sorter feeds out: 4.		
6		Pile Sorter Upgrade 6	3306	2000	2000	2000		2000	Pile Sorter loads/unloads simultaneously.		
1		Distribution Range 1[2]	4001	400	400				+20 degrees to Logistics Bot distance	+20 degrees (60 degrees total)	Requires Distribution Logistics

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
2		Distribution Range 2	4002	600 600 600	+20 degrees to Logistics Bot distance	+40 degrees (80 degrees total)	
3		Distribution Range 3	4003	800 800 800	+30 degrees to Logistics Bot distance	+70 degrees (110 degrees total)	
4		Distribution Range 4	4004	1000 1000 1000 1000	+30 degrees to Logistics Bot distance	+100 degrees (140 degrees total)	
5		Distribution Range 5	4005	1200 1200 1200 1200 1200	+40 degrees to Logistics Bot distance	+140 degrees (180 degrees total)	
1		Logistics Carrier Engine 1	3401	400 400 400	+10% logistics bot flight speed. +20% logistics drone flight speed.	+10% logistics bot flight speed (6.6 m/s). +20% logistics drone flight speed (9.6 m/s).	Requires Plane Logistics Syste
2		Logistics Carrier Engine 2	3402	600 600 600	+10% logistics bot flight speed. +20% logistics drone flight speed.	+20% logistics bot flight speed (7.2 m/s). +40% logistics drone flight speed (11.2 m/s).	
3		Logistics Carrier Engine 3	3403	800 800 800	+15% logistics bot flight speed. +30% logistics drone flight speed. +50% logistics vessel navigate speed.	+35% logistics bot flight speed (8.1 m/s). +70% logistics drone flight speed (13.6 m/s). +50% logistics vessel navigate speed (900 m/s).	Requires Interstellar Logistics Syste
4		Logistics Carrier Engine 4	3404	1000 1000 1000 1000	+15% logistics bot flight speed. +30% logistics drone flight speed. +50% logistics vessel navigate speed. Unlocks logistics vessel warp.	+50% logistics bot flight speed (9.0 m/s). +100% logistics drone flight speed (16.0 m/s). +100% logistics vessel navigate speed (1,200 m/s).	Requires Drive Engine 4.

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
										Logistics vessel warp speed: 0.10 ly/s.	
5		Logistics Carrier Engine 5	3405	1200	1200	1200	1200	1200	+15% logistics bot flight speed. +40% logistics drone flight speed. +50% logistics vessel navigate speed.	+65% logistics bot flight speed (9.9 m/s). +140% logistics drone flight speed (19.4 m/s). +150% logistics vessel navigate speed (1,500 m/s). Logistics vessel warp speed: 0.125 ly/s.	
6		Logistics Carrier Engine 6	3406	1600	1600	1600	1600	1600	+15% logistics bot flight speed. +40% logistics drone flight speed. +50% logistics vessel navigate speed.	+80% logistics bot flight speed (10.8 m/s). +180% logistics drone flight speed (21.6 m/s). +200% logistics vessel navigate speed (1,800 m/s). Logistics vessel warp speed: 0.15 ly/s.	
7		Logistics Carrier Engine ∞	3407	2000 × (LVL-6)^2					+20% logistics bot flight speed. +50% logistics drone flight speed. +50% logistics vessel navigate speed.	+(80+20*(LVL-6))% logistics bot flight speed. +(180+50*(LVL-6))% logistics drone flight speed. +(200+50*(LVL-6))% logistics vessel navigate speed. Total bot speed: (10.8+1.2*(LVL-6)) m/s Total drone speed:	Can be researc infinitely.

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
										`21.6+4*(LVL-6)` m/s. Total vessel speed: `1800+300*(LVL-6)` m/s. Total vessel warp speed: `0.15+0.025*(LVL-6)` ly/s.	
1		Logistics Carrier Capacity 1	3501	400	400	400			+1 logistics bot carrying capacity. +5 logistics drone carrying capacity.	+1 logistics bot carrying capacity (9). +5 logistics drone carrying capacity (30).	Requires Plane Logistics Syste
2		Logistics Carrier Capacity 2	3502	600	600	600			+1 logistics bot carrying capacity. +5 logistics drone carrying capacity.	+2 logistics bot carrying capacity (10). +10 logistics drone carrying capacity (35).	
3		Logistics Carrier Capacity 3	3503	800	800	800			+2 logistics bot carrying capacity. +5 logistics drone carrying capacity. +100 logistics vessel carrying capacity.	+4 logistics bot carrying capacity (12). +15 logistics drone carrying capacity (40). +100 logistics vessel carrying capacity (300).	Requires Interstellar Logistics Syste
4		Logistics Carrier Capacity 4	3504	1000	1000	1000			+2 logistics bot carrying capacity. +10 logistics drone carrying capacity. +100 logistics vessel carrying capacity.	+6 logistics bot carrying capacity (14). +25 logistics drone carrying capacity (50). +200 logistics vessel carrying capacity (400).	
5		Logistics Carrier Capacity 5	3505	1200	1200	1200	1200		+2 logistics bot carrying capacity. +10 logistics drone carrying capacity. +100 logistics vessel carrying capacity.	+8 logistics bot carrying capacity (16). +35 logistics drone carrying capacity (60). +300 logistics vessel carrying capacity (500).	
6		Logistics Carrier Capacity 6	3506	1600	1600	1600	1600		+2 logistics bot carrying capacity. +10 logistics	+10 logistics bot carrying capacity (18). +45 logistics	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
					drone carrying capacity. +100 logistics vessel carrying capacity.	drone carrying capacity (70). +400 logistics vessel carrying capacity (600).	
7		Logistics Carrier Capacity 7	3507		+2 logistics bot carrying capacity. +10 logistics drone carrying capacity. +200 logistics vessel carrying capacity.	+12 logistics bot carrying capacity (20). +55 logistics drone carrying capacity (80). +600 logistics vessel carrying capacity (800).	
8		Logistics Carrier Capacity 8	3508		+20 logistics drone carrying capacity. +200 logistics vessel carrying capacity.	+75 logistics drone carrying capacity (100). +800 logistics vessel carrying capacity (1000).	
9-11		Logistics Carrier Capacity 9-11	3509		+20 logistics drone carrying capacity. +200 logistics vessel carrying capacity. +1000 Planetary station storage caps. +2000 Interstellar station storage caps.	`100+20*(LVL-8)` logistics drone carrying capacity. `1000+200*(LVL-8)` logistics vessel carrying capacity. `5000+1000*(LVL-8)` Planetary station storage caps. `10000+2000*(LVL-8)` Interstellar station storage caps.	Each level alsc increases the storage capac of Advanced Mining Machir and Orbital Collector by 1(per level.
12-13		Logistics Carrier Capacity 12-13	3510				
1		Logistics Station Integrated Logistics 1	3801		+1 Logistics station cargo stack count	+1 Logistics station cargo stack count (2)	Requires Logis Carrier Capaci Each level of tl upgrade also increases the cargo stack cc of Advanced Mining Machir
2		Logistics Station Integrated Logistics 2	3802		+1 Logistics station cargo stack count	+2 Logistics station cargo stack count (3)	Requires Logis Carrier Capaci 11.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
3		Logistics Station Integrated Logistics 3	3803	 24.0k	+1 Logistics station cargo stack count	+3 Logistics station cargo stack count (4)	Requires Logis Carrier Capaci 13.
1		Veins Utilization 1	3601	 400  400	-6% ore consume / miner product. +10% mining speed.	-6% ore consume / miner product (94%). +10% mining speed (110%).	Requires Electromagnet
2		Veins Utilization 2	3602	 1000  1000  1000	-6% ore consume / miner product. +10% mining speed.	-11.6% ore consume / miner product (≈88.4%). +20% mining speed (120%).	
3		Veins Utilization 3	3603	 1000  1000  1000  1000	-6% ore consume / miner product. +10% mining speed.	-16.9% ore consume / miner product (≈83.1%). +30% mining speed (130%).	
4		Veins Utilization 4	3604	 1600  1600  1600  1600	-6% ore consume / miner product. +10% mining speed.	-21.9% ore consume / miner product (≈78.1%). +40% mining speed (140%).	
5		Veins Utilization 5	3605	 2000  2000  2000  2000  2000	-6% ore consume / miner product. +10% mining speed.	-26.6% ore consume / miner product (≈73.4%). +50% mining speed (150%).	
6		Veins Utilization ∞	3606	 4000 × (LVL-5)	-6% ore consume / miner product. +10% mining speed.	Total consumption ≈ 100*(0.94^LVL)%: Total speed: 100+10*LVL %	Can be researc infinitely.
1		Research Speed 1	3901	 1000	+60 Hash/s per lab researching speed.	+60 Hash/s per lab researching speed (120 Hash/s).	Requires Information Matrix.
2		Research Speed 2	3902	 2000	+60 Hash/s per lab researching speed.	+120 Hash/s per lab researching speed (180 Hash/s).	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
3		Research Speed 3	3903	4000	+60 Hash/s per lab researching speed.	+180 Hash/s per lab researching speed (240 Hash/s).	
4		Research Speed ∞	3904	8000 × (LVL-3)	+60 Hash/s per lab researching speed.	+180+60*(LVL-3) Hash/s per lab researching speed. Total: 240+60*(LVL-3) Hash/s.	Can be researc infinitely.

Combat Upgrades

These upgrades improve combat statistics such as weapon damage, attack speed, fleet size and durability.[3]

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
1		Kinetic Weapon Damage 1	5001	 100	Kinetic Weapons Damage +10%	Damage: +10% (110%)	Requires Weapon System.
2		Kinetic Weapon Damage 2	5002	 200 200	Kinetic Weapons Damage +10%	Damage: +20% (120%)	
3		Kinetic Weapon Damage 3	5003	 400 400 400	Kinetic Weapons Damage +20%	Damage: +40% (140%)	
4		Kinetic Weapon Damage 4	5004	 600 600 600 600	Kinetic Weapons Damage +30%	Damage: +70% (170%)	
5		Kinetic Weapon Damage 5	5005	 1000 1000 1000 1000 1000	Kinetic Weapons Damage +30%	Damage: +100% (200%)	
6		Kinetic Weapon Damage ∞	5006	 2000 × `(LVL-4)*(LVL-5)`	Kinetic Weapons Damage +20%	Damage: +`20*LVL`% (`100+20*LVL`%)	
1		Energy Weapon Damage 1	5101	 100	Energy Weapons Damage +10%	Damage: +10% (110%)	Requires Weapon System.
2		Energy Weapon Damage 2	5102	 200 200	Energy Weapons Damage +10%	Damage: +20% (120%)	
3		Energy Weapon Damage 3	5103	 400 400 400	Energy Weapons Damage +20%	Damage: +40% (140%)	
4		Energy Weapon Damage 4	5104	 800 800 800 800	Energy Weapons Damage +30%	Damage: +70% (170%)	
5		Energy Weapon Damage 5	5105	 2000 2000 2000 2000 2000	Energy Weapons Damage +30%	Damage: +100% (200%)	
6		Energy Weapon Damage ∞	5106	 6000 × `(LVL-5)^2`	Energy Weapons Damage +20%	Damage: +`20*LVL`% (`100+20*LVL`%)	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
1		Explosive Weapon Damage 1	5201	100	Explosive Weapons Damage +10%	Damage: +10% (110%)	Requires Combustible Unit.
2		Explosive Weapon Damage 2	5202	200, 200	Explosive Weapons Damage +10%	Damage: +20% (120%)	
3		Explosive Weapon Damage 3	5203	400, 400, 400	Explosive Weapons Damage +20%	Damage: +40% (140%)	
4		Explosive Weapon Damage 4	5204	600, 600, 600, 600	Explosive Weapons Damage +30%	Damage: +70% (170%)	
5		Explosive Weapon Damage 5	5205	1000, 1000, 1000, 1000, 1000	Explosive Weapons Damage +30%	Damage: +100% (200%)	
6		Explosive Weapon Damage ∞	5206	2000 × `(LVL-4)*(LVL-5)`	Explosive Weapons Damage +20%	Damage: +`20*LVL`% (`100+20*LVL`%)	
1		Combat Drone Damage 1	5301	300, 300	Ground Fighter Damage +10%	Ground: +10% (110%)	Requires Prototype.
2		Combat Drone Damage 2	5302	500, 500, 500	Ground Fighter Damage +10%	Ground: +20% (120%)	
3		Combat Drone Damage 3	5303	800, 800, 800, 800	Ground Fighter Damage +15% Space Fighter Damage +25%	Ground: +35% (135%) Space: +25% (125%)	
4		Combat Drone Damage 4	5304	1000, 1000, 1000, 1000, 1000	Ground Fighter Damage +15% Space Fighter Damage +25%	Ground: +50% (150%) Space: +50% (150%)	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
5		Combat Drone Damage ∞	5305	 2000 × `(3*(LVL-4)^2 - 1)`	Ground Fighter Damage +20% Space Fighter Damage +20%	Ground: +`50+20*(LVL-4)`% (`150+20*(LVL-4)`%) Space: +`50+20*(LVL-4)`% (`150+20*(LVL-4)`%)	
1		Combat Drone Attack Speed 1	5401	 200  200	Ground Fighter Attack Speed +10%	Ground: +10% (110%)	Requires Prototype.
2		Combat Drone Attack Speed 2	5402	 400  400  400	Ground Fighter Attack Speed +10%	Ground: +20% (120%)	
3		Combat Drone Attack Speed 3	5403	 800  800  800  800	Ground Fighter Attack Speed +20% Space Fighter Attack Speed +30%	Ground: +40% (140%) Space: +30% (130%)	
4		Combat Drone Attack Speed 4	5404	 1000  1000  1000  1000  1000	Ground Fighter Attack Speed +20% Space Fighter Attack Speed +30%	Ground: +60% (160%) Space: +60% (160%)	
5		Combat Drone Attack Speed 5	5405	 1600	Ground Fighter Attack Speed +40% Space Fighter Attack Speed +40%	Ground: +100% (200%) Space: +100% (200%)	
1		Combat Drone Durability 1	5601	 200  200	Fighter Durability +10%	Fighter: +10% (110%)	Requires Prototype.

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
2		Combat Drone Durability 2	5602	400 400 400	Fighter Durability +10%	Fighter: +20% (120%)	
3		Combat Drone Durability 3	5603	600 600 600 600	Fighter Durability +20% Warship Durability +30%	Fighter: +40% (140%) Warship: +30% (130%)	
4		Combat Drone Durability 4	5604	1000 1000 1000 1000 1000	Fighter Durability +20% Warship Durability +30%	Fighter: +60% (160%) Warship: +60% (160%)	
5		Combat Drone Durability 5	5605	2000	Fighter Durability +40% Warship Durability +40%	Fighter: +100% (200%) Warship: +100% (200%)	
1		Ground Squadron Expansion 1	5801	200 200	Ground Squadron Slots +2	Slots: +2 (6)	Requires Prototype.
2		Ground Squadron Expansion 2	5802	600 600 600	Ground Squadron Slots +2	Slots: +4 (8)	
3		Ground Squadron Expansion 3	5803	1600 1600 1600 1600	Number of Ground Squadrons +1	Slots: +4 (8) Squadrons: +1 (2)	
4		Ground Squadron Expansion 4	5804	2400 2400 2400 2400	Ground Squadron Slots +4	Slots: +8 (12) Squadrons: +1 (2)	
5		Ground Squadron Expansion 5	5805	4000 4000 4000 4000 4000	Number of Ground Squadrons +2	Slots: +8 (12) Squadrons: +3 (4)	
6		Ground Squadron Expansion 6	5806	6000	Number of Ground Squadrons +2	Slots: +8 (12) Squadrons: +5 (6)	
7		Ground Squadron Expansion 7	5807	8000	Number of Ground Squadrons +2	Slots: +8 (12) Squadrons: +7 (8)	

Tier	Icon	Upgrade	ID	Resource Consumption	Unlocks	Cumulative Effect (Total)	Notes
1		Space Fleet Expansion 1	5901	400 400 400 400	Space Fleet Slots +4	TBD	Requires Corvette.
2		Space Fleet Expansion 2	5902	800 800 800 800	Space Fleet Slots +4	Slots: +4 (8)	
3		Space Fleet Expansion 3	5903	1800 1800 1800 1800	Number of Space Fleets +1	Slots: +4 (8) Fleets: +1 (2)	
4		Space Fleet Expansion 4	5904	4000 4000 4000 4000 4000	Number of Space Fleets +1	Slots: +4 (8) Fleets: +2 (3)	
5		Space Fleet Expansion 5	5905	6000 6000 6000 6000 6000	Number of Space Fleets +1	Slots: +4 (8) Fleets: +3 (4)	
6		Space Fleet Expansion 6	5906	8000	Number of Space Fleets +2	Slots: +4 (8) Fleets: +5 (6)	
7		Space Fleet Expansion 7	5907	12.0k	Number of Space Fleets +2	Slots: +4 (8) Fleets: +7 (8)	
1		Enhanced Structure 1[4]	6001	200	Durability of every unit +4%	Durability: +4% (104%)	Requires Mecha Core 1.
2		Enhanced Structure 2	6002	400 400	Durability of every unit +4%	Durability: +8% (108%)	
3		Enhanced Structure 3	6003	600 600 600	Durability of every unit +4%	Durability: +12% (112%)	
4		Enhanced Structure 4	6004	1000 1000 1000 1000	Durability of every unit +4%	Durability: +16% (116%)	
5		Enhanced Structure 5	6005	1500 1500 1500 1500 1500	Durability of every unit +4%	Durability: +20% (120%)	
6		Enhanced Structure ∞	6006	500 × `(LVL-5)^2+3*(LVL-5)`	Durability of every unit +4%	Durability: + `4 * LVL` % (`100+4*LVL` %)	

Tier	Icon	Upgrade	ID	Resource Consumption					Unlocks	Cumulative Effect (Total)	Notes
1		Planetary Shield 1	5701	200	200	200			Planet Shield Energy Field Eff. +10%	Efficiency: +10% (110%)	Requires Planetary Defense System.
2		Planetary Shield 2	5702	400	400	400			Planet Shield Energy Field Eff. +15%	Efficiency: +25% (125%)	
3		Planetary Shield 3	5703	600	600	600	600		Planet Shield Energy Field Eff. +20%	Efficiency: +45% (145%)	
4		Planetary Shield 4	5704	1000	1000	1000	1000	1000	Planet Shield Energy Field Eff. +25%	Efficiency: +70% (170%)	
5		Planetary Shield 5	5705	2000					Planet Shield Energy Field Eff. +30%	Efficiency: +100% (200%)	

Player Tips

- For the upgrades that can be researched infinitely, their research consumption increases each time it is researched. This leads to them being considerably more expensive to keep researching, the further you research them.
- It is worth trying for some of the upgrades as soon as possible. The following recommended upgrades do not use matrices or technology unlocks, and can therefore be researched immediately upon starting the game:
  - Drive Engine 1 (allows sustained flight, which is faster than walking, but consumes more energy)
  - Mecha Core 1 (increases Icarus's maximum energy storage)
  - Universe Exploration 1 (displays symbols for resources and their yields across the planet)
- Energy Circuit 1 is also extremely useful in the early game for recharging Icarus faster, but requires the Smelting Purification technology to produce required resources.

Notes

1. Added in the Patch 0.10.28.20729, December 15th, 2023
2. Added in the Patch 0.9.27.14546, September 27th, 2022
3. Unless otherwise noted, combat upgrades have been added in the Patch 0.10.28.20729, December 15th, 2023
4. Added in the Patch 0.10.28.20904, December 18th, 2023

Research: Technologies and Upgrades

Technologies

Tiers 0-1

Dyson Sphere Program · Electromagnetism

Tier 2

Basic Logistics System · Automatic Metallurgy · Electromagnetic Matrix · Basic Assembling Processes · Fluid Storage Encapsulation · High-Efficiency Plasma Control · Electromagnetic Drive · Engine · Weapon System

Tier 3

Improved Logistics System · Steel Smelting · Combustible Unit · Smelting Purification · Thermal Power · Plasma Extract Refining · Battlefield Analysis Base

Tier 4

Environment Modification · Crystal Smelting · Solar Collection · Semiconductor Material · Deuterium Fractionation · Proliferator Mk.I · Basic Chemical Engineering · Energy Matrix · Magnetic Levitation · Missile Turret · Prototype

Tier 5

High-Efficiency Logistics System · Titanium Smelting · Energy Storage · Photon Frequency Conversion · Geothermal Extraction · Processor · Applied Superconductor · Polymer Chemical Engineering · X-Ray Cracking · Hydrogen Fuel Rod · Super Magnetic Field Generator · Signal Tower · Implosion Cannon · Precision Drone

Tier 6

Integrated Logistics System · Distribution Logistics System · Planetary Logistics System · Solar Sail Orbit System · High-Speed Assembling Processes · High-Strength Crystal · Reformed Refinement · Thruster · Proliferator Mk.II · Magnetic Particle Trap · Titanium Ammo Box

Tier 7

High-Strength Titanium Alloy · High-Strength Lightweight Structure · Ray Receiver · Mini Fusion Power Generation · High-Strength Material · Structure Matrix · Explosive Unit · Reinforced Thruster · High-Explosive Shell Set · Planetary Defense System · Superalloy Ammo Box · Attack Drone

Tier 8

Interstellar Logistics System · Interstellar Power Transmission · Particle Control · High-Strength Glass · Casimir Crystal · Miniature Particle Collider · Proliferator Mk.III · Satellite Power Distribution System · Supersonic Missile Set

Tiers 9-10

Gas Giants Exploitation · Information Matrix · Wave Function Interference · Crystal Explosive Unit · Strange Matter · Crystal Shell Set · Plasma Turret · Corvette · Vertical Launching Silo · Quantum Chip · Plane-Filter Smelting · Gravitational Wave Refraction · *Digital Analog Computation*

Tiers 11-16

Dyson Sphere Stress System · Planetary Ionosphere Utilization · Quantum Printing · Photon Spotlight Mining · Gravity Matrix · Gravity Missile Set · Destroyer · Dirac Inversion Mechanism · Controlled Annihilation Reaction · Artificial Star · Antimatter Capsule · Universe Matrix · **Mission Completed!** · *Negentropy Recursion* · *Matter Recombination* · *High-Density Controlling Annihilation*

Upgrades

Exploration

Universe Exploration

Mecha

Mecha Core · Mechanical Frame · Inventory Capacity · Auto Reconstruction Marking · Communication Control · Mass Construction · Energy Circuit · Drone Engine · Drive Engine · Energy Shield

Building

Solar Sail Life · Solar Sail Attaching Speed · Ray Transmission Efficiency · Vertical Construction · Pile Sorter Upgrade · Distribution Range · Logistics Carrier Engine · Logistics Carrier Capacity · Logistics Station Integrated Logistics · Veins Utilization · Research Speed

Combat

Kinetic Weapon Damage · Energy Weapon Damage · Explosive Weapon Damage · Combat Drone Damage · Combat Drone Attack Speed · Combat Drone Durability · Ground Squadron Expansion · Space Fleet Expansion · Enhanced Structure · Planetary Shield

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